

THE POLYMATH INITIATIVE

TRACK 1 — AI & ENGINEERING

EXPERIMENTAL PROTOCOL LOCK v1.0

Decisions and specifications discussed for the core experiment

1. Research Question

For open-weight language models $\leq 14\text{B}$ parameters, how do different memory architectures — varying in what they store and how they structure and retrieve it — affect factual retention and downstream task accuracy across extended multi-session conversations, and how does this interact with model size within the $\leq 14\text{B}$ range?

2. The Six Locks

#	Area	Protocol decision
1	Final model registry	Llama 3.2 3B / 8B / 11B + Gemma 3 4B / 12B
2	Inference	Greedy decoding; temperature 0; top-p 1.0; no sampling
3	Memory architectures	No memory + Recency + Semantic + Summary + Hybrid
4	Retrieval	all-MiniLM-L6-v2; cosine similarity; top-k = 5; no reranker; atomic memory units
5	Metrics	Factual Retention Accuracy + downstream task accuracy + session-distance analysis
6	Validation	Automated scoring $\rightarrow$ 3–5 human judges $\rightarrow$ Krippendorff's α $\rightarrow$ validated LLM judge where needed

3. Final Model Registry

Primary family — Llama 3.2:

- Llama 3.2 3B — small model.
- Llama 3.2 8B — medium model.
- Llama 3.2 11B Vision, used in text-only mode — large model.

Cross-family validation — Gemma 3:

- Gemma 3 4B.
- Gemma 3 12B.

DeepSeek-R1-Distill variants are not part of the primary statistical comparison because their distilled-model status introduces an additional training-history consideration. Qwen is not a mandatory model because the core experiment should not depend on an unreliable API pipeline.

Limitation: Llama 3.2 11B Vision is not a perfectly clean text-only scaling continuation of the 3B/8B models. Results should therefore not be interpreted as if parameter count were the only factor changing.

4. Inference Method — LOCKED

Primary experiment: deterministic greedy decoding.

- temperature = 0
- top_p = 1.0
- do_sample = False where supported
- No sampling-based variation in the primary experiment.

Each run should also record the model/checkpoint identifier, tokenizer, inference library/version, quantization, hardware, prompt/system templates, generation settings, seed where applicable, and run ID.

5. Memory Architectures — LOCKED

- A — No External Memory (control): only the defined conversation context is available.
- B — Recency / Episodic Memory: recent conversational information is stored as raw memory entries and selected using a recency-first policy.
- C — Semantic Vector Memory: atomic memory units are embedded and retrieved by semantic similarity.
- D — Summarized Memory: accumulated information is periodically compressed into a structured summary used as memory.
- E — Hybrid Memory: combines recent episodic memory, semantic retrieval, and structured summary.

These five conditions provide a controlled progression from no external memory to recency, semantic retrieval, compression, and a combined architecture.

6. Retrieval Configuration — LOCKED

- Embedding model: sentence-transformers/all-MiniLM-L6-v2.
- Similarity: cosine similarity.
- Memory units: atomic facts/events/instructions rather than arbitrary fixed-length document chunks.
- Top-k: 5.
- Reranking: disabled in the primary experiment.
- Tie-breaking: similarity score descending → recency → stable memory ID.

The same retrieval configuration should be used across models in the primary comparison. Any change caused by technical feasibility must be documented as a protocol amendment.

7. Conversation Horizons — LOCKED

- 10 simulated sessions — short horizon.
- 50 simulated sessions — medium horizon.
- 100 simulated sessions — long horizon.

A session should contain conversational interaction, introduction or reinforcement of facts, appropriate distractor material, and later evaluation probes. Evaluation probes should be held out from the preceding conversation rather than simply repeating the exact training question.

8. Factual Retention — LOCKED

Primary metric: Factual Retention Accuracy (FRA).

$\text{FRA} = \text{correctly retained facts} / \text{total facts probed}$.

Example: 73 correct answers out of 100 probed facts = 73% FRA.

Also report retention by distance from fact introduction:

- 1–5 sessions
- 6–10 sessions
- 11–25 sessions
- 26–50 sessions
- 51–100 sessions

9. Downstream Task Accuracy — LOCKED

Use tasks that test whether retained information remains useful.

- Factual QA — requires previously introduced information.
- Multi-fact reasoning — requires combining two or more stored facts.
- State-dependent task — correct output depends on remembering an earlier user state, preference, or instruction.

For objectively scorable tasks: $\text{Accuracy} = \text{correct responses} / \text{total responses}$.

Report category-level accuracy and aggregate downstream-task accuracy.

10. Experimental Factors

Independent variables:

- Model family: Llama vs Gemma.
- Model size: 3B, 4B, 8B, 11B, 12B.
- Memory architecture: five conditions.
- Conversation horizon: 10, 50, 100 sessions.

Primary dependent variables:

- Factual retention accuracy.
- Downstream task accuracy.

Secondary measurements may include retention by session distance, retrieval hit rate/precision where ground truth permits, inference latency, and memory storage footprint.

11. Judge / Human Validation Protocol — LOCKED

1. Use automated scoring whenever an objective ground-truth answer exists.
2. Select a stratified representative sample covering model family, model size, memory architecture, session horizon, and task type.
3. Have 3–5 independent human annotators score the sample using the same rubric.
4. Measure inter-rater agreement using Krippendorff's α with the appropriate measurement level.
5. Only after human validation should an LLM judge be used for subjective outputs at scale.
6. Treat the validated LLM judge as a scalable secondary evaluator, not the ultimate ground truth.

The current Track 1 proposal identifies GPT-OSS 120B and Llama 3.3 70B as possible judge models. Their exact use remains an implementation choice after the human rubric/validation stage.

12. The Existing Memory-Decay Code

The existing Python implementation demonstrates two concepts: an exponential forgetting curve and interference/catastrophic forgetting. It is useful as a conceptual baseline, but it does not yet implement real LLMs, memory architectures, retrieval, multi-session conversations, or the protocol's retention/downstream metrics.

Recommended label: “Track 1 — Conceptual Memory Decay / Interference Demonstration.”

13. Actual Experimental Pipeline

Conversation Generator

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Memory Writer

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Memory Store

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Memory Retrieval

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Open-weight LLM

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Response

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Evaluation

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Raw CSV + experiment metadata

14. What We Are Explicitly Not Doing

- No manual chat-interface evaluation as the primary method.
- No arbitrary API switching halfway through main data collection.
- No sampling in the primary experiment.
- No model-specific retrieval settings in the primary comparison.
- No reranking in the primary experiment.
- No DeepSeek variants in the primary statistical comparison.
- No presenting the conceptual forgetting code as the actual LLM experiment.

15. Core Experimental Matrix

Model	No memory	Recency	Semantic	Summary / Hybrid
Llama 3.2 3B	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions
Llama 3.2 8B	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions
Llama 3.2 11B	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions
Gemma 3 4B	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions
Gemma 3 12B	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions	10 / 50 / 100 sessions

This table shows the core factorial structure. The exact number of independent repetitions/seeds must still be established before the full run so that statistical uncertainty can be estimated.

16. Feasibility Gate Before Full Runs

7. Verify every candidate model can be loaded and run reliably.
8. Benchmark a small number of sessions before committing to the full matrix.
9. Measure loading time, inference time, resource usage, stability, and reproducibility.
10. Confirm that the pipeline can produce the required raw CSV and metadata.
11. Only then begin the full 50- and 100-session experiments.

If a model proves technically infeasible, the registry may be amended before main data collection. The amendment must be documented rather than silently changing the experiment.

17. Immediate Next Step — Minimum Viable Experiment

With the six protocol areas locked, Track 1 should build the MVE: one model + one memory architecture + fixed inference/retrieval settings + a small conversation set + automated evaluation + CSV output.

The MVE's purpose is to prove that the entire pipeline works before scaling to 50- and 100-session runs.

18. Final Principle

We are not choosing the easiest models or APIs. We are building a controlled and reproducible experiment in which memory architecture and model size can be meaningfully compared.